Example 6.14

Calculate the price index and quantity index for the following data by Fisher's ideal formula and verify that it statisfies the factor reversal test.

Commodity	Base year		Current year	
	Price (₹)	Quantity (`000 tonnes)	Price (₹)	Quantity (`000 tonnes)
A	40	70	40	32
B	50	84	30	80
C	60	58	25	50

Solution

Commodity	p_0	q_0	p_1	q_1	p_1q_1	p_1q_0	p_0q_0	p_0q_1
A	40	70	40	32	1280	2800	2800	1280
B	50	84	30	80	2400	2520	4200	4000
C	60	58	25	50	1250	1450	3480	3000
					5930	6770	10480	8280

Factor Reversal test: $P_{01} \times Q_{01} = \frac{\sum p_1q_1}{\sum p_0q_0}$

$$P_{01} = \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}}$$

$$Q_{01} = \sqrt{\frac{\sum q_1p_0}{\sum q_0p_0} \times \frac{\sum q_1p_1}{\sum q_0p_1}}$$

$$P_{01} \times Q_{01} = \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1} \times \frac{\sum q_1p_0}{\sum q_0p_0} \times \frac{\sum q_1p_1}{\sum q_0p_1}}$$

$$= \sqrt{\frac{6770}{10480} \times \frac{5930}{8280} \times \frac{8280}{10480} \times \frac{5930}{6770}}$$

$$= \left(\sqrt{\frac{5930}{10480}}\right)^2$$

$$= \frac{5930}{10480}$$

$$= \frac{\sum p_1q_1}{\sum p_0q_0}$$

Hence, Fisher ideal index number satisfies the factor reversal test

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